

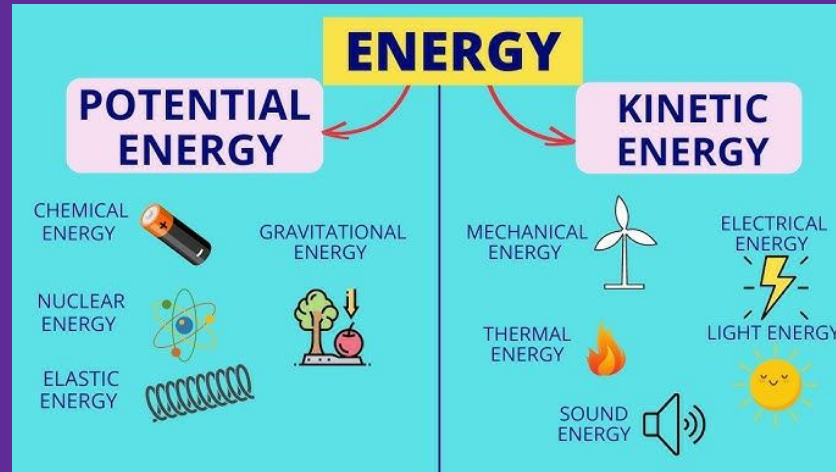
ENERGY AND SYSTEMS



WHAT IS ENERGY?

In physics, energy is the quantitative property that is transferred to or from a body or a physical system, recognizable in the performance of work. (*We will discuss work later in the unit.*)

Energy is measured in Joules, J. It is a scalar. It comes in different forms. Let's look some types of energy.



Types of energy

□ Kinetic Energy

- Energy due to motion
- $K = \frac{1}{2} m v^2$
 - K: Kinetic Energy in J
 - m: mass in kg
 - v: speed in m/s
- Unit: Joules
- Kinetic energy is independent of the direction of motion; therefore it is a scalar quantity.
- Kinetic energy is always positive.

Sample problem

A 10.0 g bullet is moving right at a speed of 1.2 km/s.

a) What is the kinetic energy of the bullet?

7200 J

b) What is the bullet's kinetic energy if it is moving at the same speed 20° above the horizontal?

7200 J

c) What is the bullet's kinetic energy if it is moving at the same speed 60° south of west?

7200 J

Potential energy

- Energy of position or configuration
- “Stored” energy
- For gravity: $U_g = mgh$ (gravitational potential energy)
 - m : mass, kg
 - g : acceleration due to gravity, m/s/s
 - h : height above the “zero” point, m (The zero point is arbitrary. You can choose its location although there are better choices for certain problems.)

When the earth or some really massive object (like a planet or moon, etc) is not in the defined system, assume that there is no gravitational potential energy in the system.

Universal gravitational potential energy

$$U_g = -Gm_1m_2/r$$

This shows that it takes at least two objects for a system to have gravitational potential energy.

The negative sign shows that the objects are bound to one another. It also shows that the gravitational potential energy increases as the separation between the masses increases.

When it comes to the earth or larger bodies compared to a much smaller mass the expression becomes:

$$U_g = -GMm/r$$

Elastic potential energy

- Ideal springs obey Hooke's law

$$\square F_s = -kx$$

- F_s - spring force, N
- k : spring force constant, N/m (*stiffness of the spring*)
- x : displacement from equilibrium position, m

- For springs: $U_s = \frac{1}{2} k x^2$ (elastic potential energy)
- U_s = elastic potential energy (J, joules)

MECHANICAL ENERGY

$$\mathbf{E \text{ or } TME = KE + U_g + U_s}$$

E - mechanical energy

TME - total mechanical energy

Mechanical energy and total energy are not the same thing.

$$\mathbf{Total \ Energy = KE + U_g + U_s + E_i}$$

E_i - internal energy such as: chemical potential energy, thermal energy, or nuclear energy

Open and closed systems

[OPEN AND CLOSED SYSTEMS LINK](#)

Law of Conservation of Mechanical Energy

- $E = K + U = \text{Constant}$

- K: Kinetic Energy ($1/2 mv^2$)

- U: Potential Energy (gravity or spring)

- $\Delta E = \Delta U + \Delta K = 0$

- ΔK : Change in kinetic energy

- ΔU : Change in gravitational or spring potential energy